IEEE OJCOMS • SPECIAL ISSUE: DELAY-TOLERANT NETWORKING FOR 6G

Peak Age of Information over Scheduled, Energy-Constrained Multi-Segment DTNs with Bundle Replication

When is a bundle copy worth its energy? A freshness theory for LEO + UAV + terrestrial contact plans.

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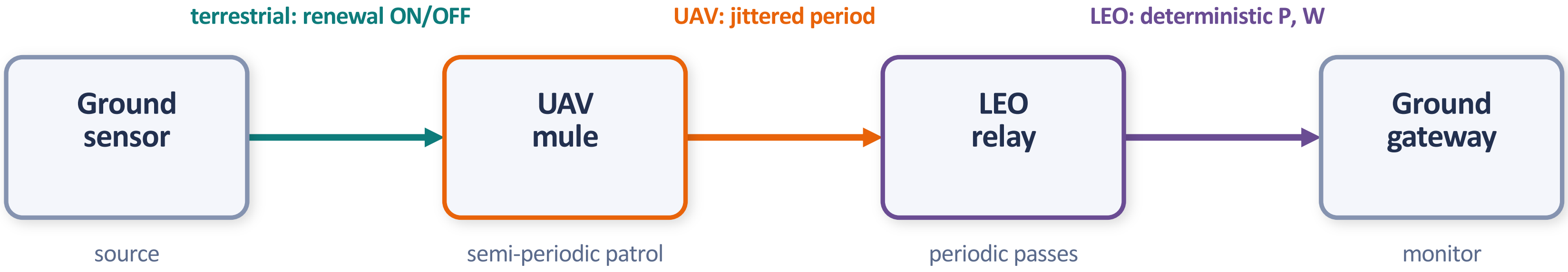
THE SETTING

Scheduled, intermittent connectivity is the 6G frontier



Links are scheduled, not lossy

A ground station sees a LEO satellite for a window W once per period P. Between passes the link is OFF by schedule — bundles age in a buffer, not in a queue.

Freshness is the metric

Telemetry, disaster response, IoT monitoring care about Age of Information and its peak (PAoI) — not throughput or delivery ratio.

Copies cost energy

DTN replication hedges uncertain contacts, but on harvesting relays every copy drains the battery that future contacts need.

POSITIONING

Three literatures that don't meet

AoI over erasure links

Chiariotti et al., HARQ / satellite AoI

Always-on server, random packet erasure. No scheduled OFF periods — no residual-wait law.

Tandem-queue AoI

Sinha et al., Senthilkumar et al.

Persistent or memorylessly failing servers. A contact plan is deterministic-periodic — a different law.

Contact-plan routing

CGR / RUCoP / CGR-UCoP

Optimizes delivery ratio under uncertain contacts. No freshness objective, no energy budget.

The gap: no prior work couples **scheduled store–carry–forward gating** + **multi-copy replication** + **an energy constraint** with a freshness objective.

Closest late-breaking work — Badia et al. (Globecom-W '25): copy diversity lowers AoI under *random* intermittency, energy-agnostic. We cover the complementary regime: deterministic contact plans, finite-battery copy-degree control, CGR realization.

Duty-cycled segments, buffered aging, an energy gate

$$R_s$$

residual wait until next window at segment s

$$D_s = R_s + T_s$$

per-segment increment (wait + service/propagation)

$$Y = \Sigma_s D_s$$

end-to-end latency; k copies → $Y_{min} = \min_k Y^{(k)}$

$$\mathrm{PAoI} = E[Z] + E[Y_{min}]$$

per-outage peak age (worst staleness per silent gap)

Energy gate. Each relay has battery b_s, harvest rate λ_e, per-copy cost e. A contact is usable only if the battery covers the copy: $I_s^{eff} = I_s \cdot \mathbb{1}\{b_s \geq e\}$. Scarcity thins contacts; replication drains the battery faster — the tension this paper resolves.

RESULT 1

The LEO residual-wait anchor — closed form

A stationary arrival meets the schedule at uniform phase:

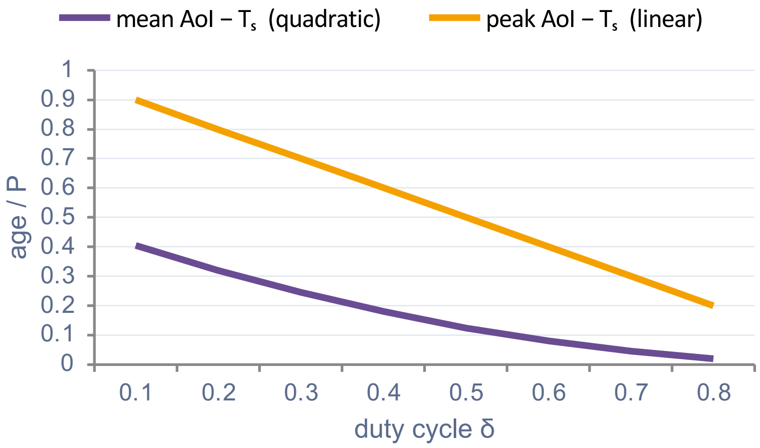
$$P(R = 0) = \delta \qquad f_r(r) = 1/P \text{ on } (0, G]$$

$$E[R] = (P-W)^2 / 2P = (P/2)(1-\delta)^2$$

$$\text{mean AoI} = T_s + (P/2)(1-\delta)^2$$

$$\text{peak AoI} = T_s + P(1-\delta)$$

Validated: KS distance 0.002; log–log slopes 1.98 / 1.00.



Design consequence: sparse contacts hurt the mean quadratically but the peak only linearly — mean-optimal and peak-optimal contact provisioning ***differ***.

Renewal generalization & the AoI/PAoI inversion

Alternating renewal: ON duration U, OFF gap V;
cycle C = U + V

$$E[R] = E[V^2] / 2E[C]$$

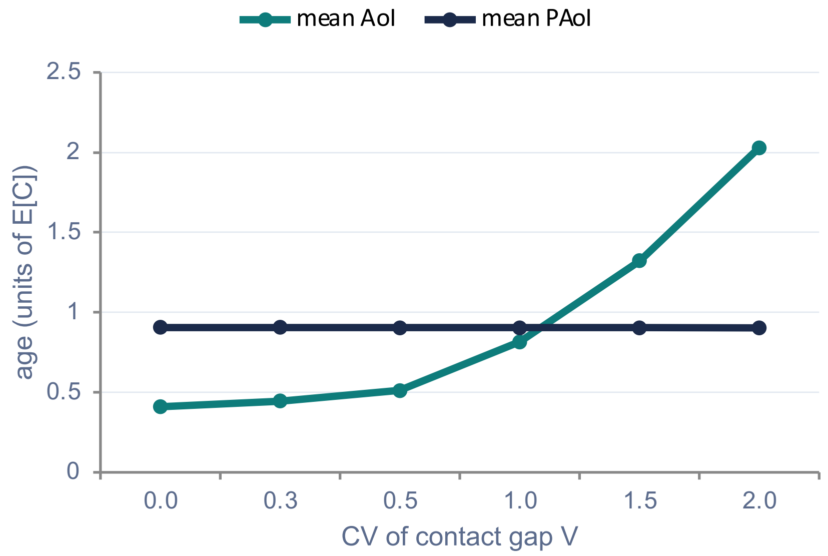
$$E[PAoI] = T_s + E[V]$$

LEO = deterministic V • UAV = low-jitter V • CTMC = exponential V • terrestrial = high-variance V

Exact inversion condition

$$CV^2(V) > 1 + 2E[U]/E[V]$$

→ peak age falls below average age (≈ CV 1.11 here)



simulation, 12 seeds — PAoI is variance-blind; AoI is not. They cross at the predicted threshold.

RESULT 2

Phase mixing composes latencies — and prices a copy

Phase-mixing lemma (Weyl)

If segment periods are incommensurate (or upstream dispersion exceeds P_s), arrival phases become uniform & independent ⇒ increments D_s are independent and end-to-end latency is their convolution.

Two-copy PAoI theorem

$$E[Y_{min}] = E[Y] - \tfrac{1}{2}\, E|Y^{(1)} - Y^{(2)}|$$

The freshness bought by a second copy = half the mean absolute dispersion of path latencies (a Gini-type index).

Failure mode: a UAV synchronized to one specific LEO pass locks the phases — the correction term revives. Confirmed in dtnsim: end-to-end variance additivity holds under incommensurate periods and collapses under commensurate (locked) ones.

$\tfrac{2}{3}(1-\delta)$

two-copy residual ratio $E[R_{min}]/E[R]$ — a 33% cut at sparse contacts

0

gain for same-bottleneck copies — replication pays only on independent contacts

RESULT 3

The energy–replication threshold

Mean-PAoI objective in copies k :

$$A(k) = P(1-p_e)/p_e + P(1-\delta)^{k+1}/(k+1)$$

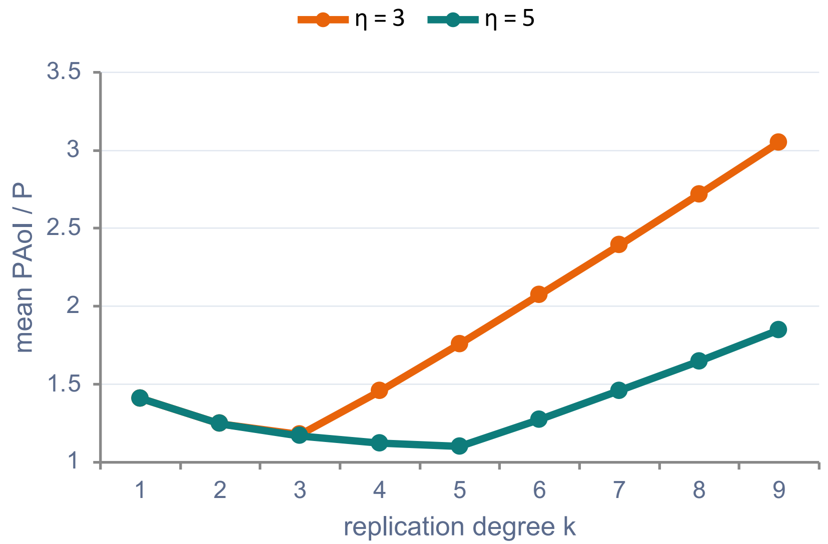
starvation (energy-forced skips) vs **order-statistic gain**

$\eta = \lambda_e P/e$ (copies harvested per contact period)

Unimodal ⇒ optimum brackets η

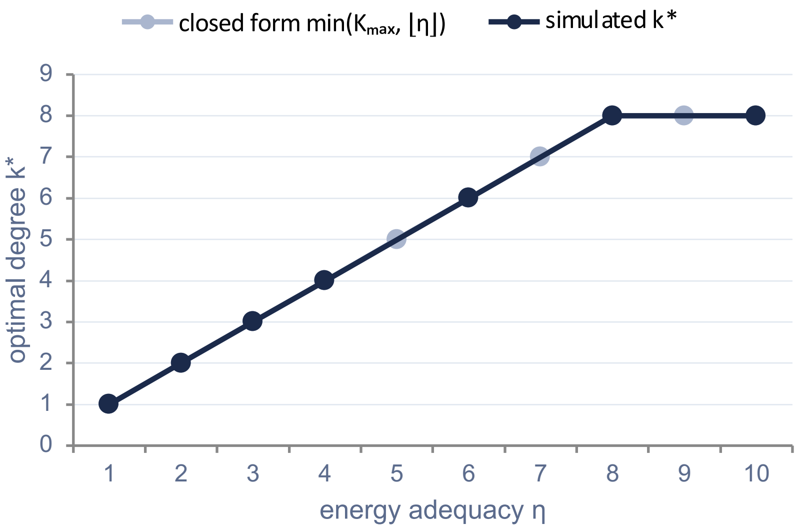
$k^* = \min(K_{max}, \lfloor\eta\rfloor \text{ or } \lceil\eta\rceil)$

$\lfloor\eta\rfloor$ = conservative no-starvation rule; take $\lceil\eta\rceil$ only when the extra copy repays its skips. Monotone in η, saturates at K_{max}.



simulated PAoI is unimodal with minimum at k = η — over-replication starves future windows.

The staircase, and the ceiling branch caught in the act



k(η) non-decreasing, saturates at K_{max} = 8 (Theorem 3)*

The ceiling branch (E14)

Fine η-sweep across [3, 4], δ = 0.1, 16 seeds:

η	3.1 – 3.8	3.9
simulated optimum	k = 3 = ⌊η⌋	k = 4 = ⌈η⌉
analytic branch	floor	ceiling

Flip lands exactly at the analytic crossover **η★ = 3.824** — 9/9 points match. Occasional energy-forced skips are worth the extra copy near the top of the interval.

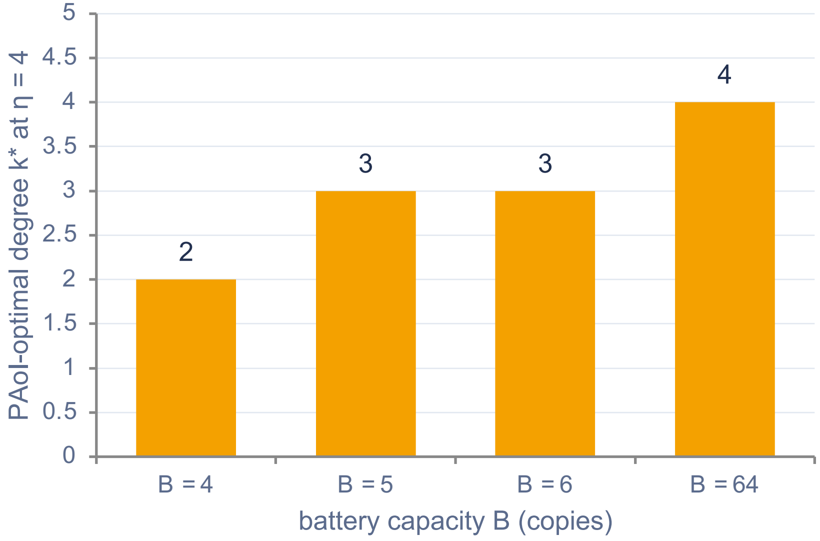
An exact battery-queue throttle, not a fluid guess

Work conservation (distribution-free)

$$k \cdot p_e(k) = \eta - L$$

$L \geq 0$ = harvest lost to the battery cap. Only the mean η enters; the fluid rule $p_e = \min(1, \eta/k)$ is exactly the infinite-battery limit. Holds to 10^{-5} in simulation, Poisson and deterministic harvest alike.

Finite battery ⇒ downward pressure on k*, not a hard ceiling: the overflow loss $L(B,k)$ varies with k, so the p_e bound doesn't order the objectives — a ceiling winner keeps winning for large enough B.



shrinking the battery pushes k down (at η = 6 it falls 6 → 4 by B = 6) — quantified, not assumed.*

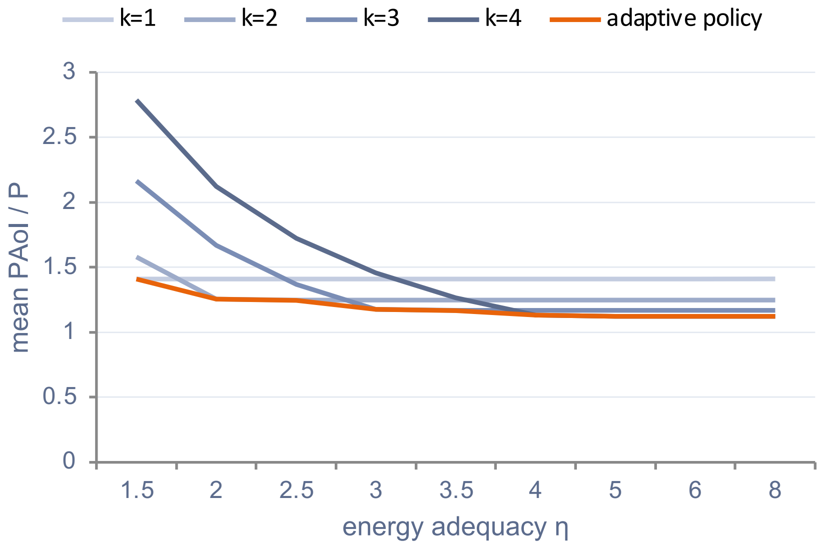
An O(1) two-candidate policy on top of CGR

The rule

k_{pol} = argmin over { ⌊η⌋, ⌈η⌉ } of A(k), clipped to [1, K_{max}]

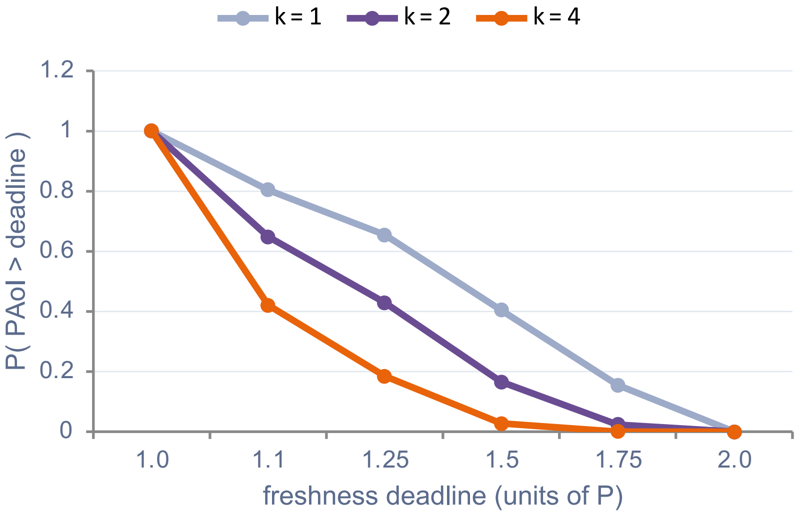
Two objective evaluations — O(1), local, no sweep. Driven by the slow mean energy-adequacy η (a set-point, not a per-window controller). CGR picks the routes; the policy picks how many.

Prediction-error robustness: as contact predictions degrade (α → 0.4), the optimum stays at ⌊η⌋ (+19% PAoI); the "reserve energy" heuristic under-replicates (+50%). Replication is the right hedge.



the policy (orange) rides the lower envelope of every fixed degree — matched the exhaustive optimum at all η.

Deadline tails collapse; RUCoP composes, not competes



15× fewer 1.5P-deadline violations at k = 4; p99 drops 1.90P → 1.59P.

vs. a RUCoP-style MDP core

Holder-set backward induction (delivery-optimal, energy-blind) on an illustrative uncertain diamond. The delivery-vs-copies frontier is concave — copies past the knee buy little.

94.7%

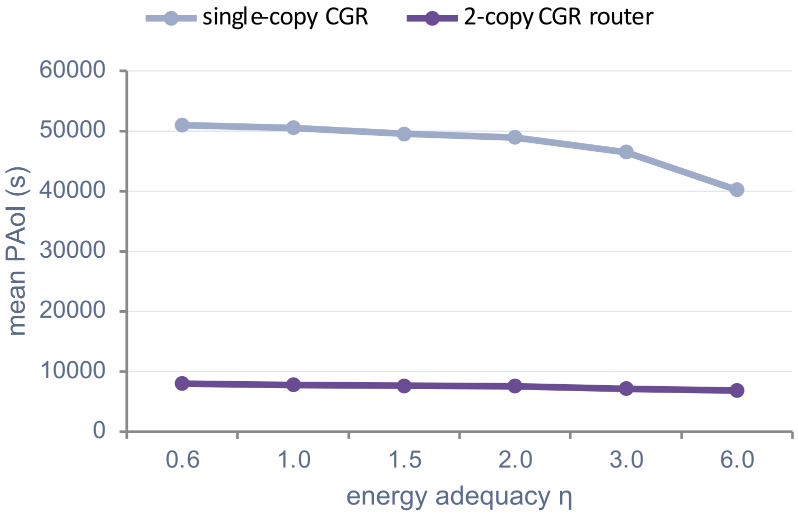
of the MDP's delivery at our k∗ = 2

40%

of its energy spend

Complementary: the MDP picks which routes, our threshold picks how many copies — the natural energy-aware extension of CGR-UCoP.

Analysis → Monte-Carlo → real Bundle Protocol/CGR



dtnsim (real BP/CGR stack), 10 seeds, 95% CIs — two copies cut PAoI ~6× by hedging relay faults.

0.0%

delay error vs closed form (2453.9 s vs 2453.9 s, full mixed residual histogram)

0.06%

error on a real Iridium-NEXT TLE contact plan (101 passes, Skyfield)

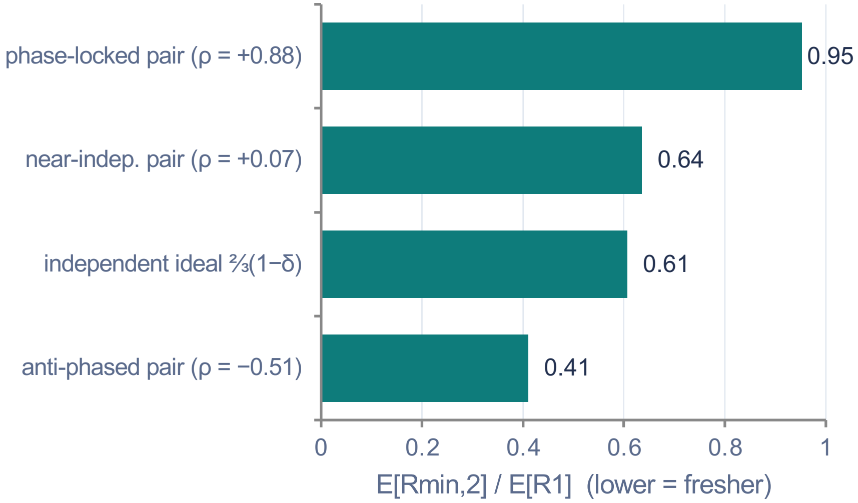
1.1%

additive-residual error on the 3-hop heterogeneous chain (sensor→UAV→LEO→gateway)

14 / 14

experiments pass (E1–E14, incl. the η★ = 3.82 ceiling flip)

Real ephemeris and flight software close the loop



44 real Iridium-NEXT pairs over Svalbard: the gain is set by pass-phase alignment — plannable from the contact plan → phase-aware copy placement.

ION 4.1.4 — reference BP flight software

974/974	24.34 s	2.5 ± 0.5 s
bundles delivered lossless, 2 h run	residual, exact per bundle (law: 24.30 s)	dispatch lag = the model's T_s, measured

delay = R + T_s decomposition validated in flight code; T_s ≈ 0.04% at operational periods.

Bursty harvest (E15)

k* = ⌊η⌋ unchanged up to harvest dwell ≈ 10 periods; work conservation ≤ 3×10^{-4} under all correlation (distribution-free). Correlation moves the **tail** (p99 +72%), not the threshold; boundary at dwell ≈ 50.

TAKEAWAYS

Freshness has a price in copies — now we know it

A residual-wait law, not a queueing delay

One alternating-renewal model covers LEO / UAV / terrestrial; mean quadratic, peak linear — and peak can drop below mean past an exact CV threshold.

Replication is priced by dispersion

A second copy buys ½ $E|Y^{(1)}-Y^{(2)}|$ of freshness — phase mixing tells you exactly when latencies compose.

k∗ tracks the energy budget

Launch ⌊η⌋ copies — or ⌈η⌉ when a skip is worth it — capped at K_{max}. An O(1) rule that rides the optimum on top of CGR.

Validated where DTN insiders live

Model-exact MC + real BP/CGR stack + ION flight software + real Iridium ephemeris — where phase-aware copy placement emerges as a plannable design rule.

Ongoing: regime-aware adaptation under extreme harvest persistence • RUCoP's route set under the degree cap • tail-constrained policy